

P01 - Reliable Batch Pipeline

CSV + paginated API + operational database | raw -> validate/quarantine -> trusted -> evidence

Business/domain

ParcelMarket order fulfillment

How this book works

This book has exactly 12 phase pages. Each phase has one artifact/evidence contract and one protected rubric page. The small teaching pattern is not the final solution; apply the reasoning to the supplied project workspace yourself.

Project control before Phase 1

- Keep protected Review and sealed failure/incident packs closed until the named phase.
- Create a real project branch/working copy if using Git.
- Record tool/runtime versions only after observing them.
- Do not delete a bad first attempt.
- Use the exact phase evidence ID and path; this keeps Mentor/book/evidence routing one-to-one.

Phase map

P01-01 - Read the batch-pipeline brief and define business questions

P01-02 - Profile every source; state grain, keys and defects

P01-03 - Draw architecture and create the repository scaffold

P01-04 - Preserve raw landing evidence and source/run metadata

P01-05 - Build ingestion with validation and quarantine

P01-06 - Add incremental state and rerun-safe identity

P01-07 - Build trusted PostgreSQL/dbt-style transformations

P01-08 - Add tests and source-to-target reconciliation

P01-09 - Automate the repeatable run/orchestration path

P01-10 - Inject failure, diagnose, repair and prove recovery

P01-11 - Finish README, runbook, architecture and evidence index

P01-12 - Clean-clone rerun and project defense

P01-01 - Read the batch-pipeline brief and define business questions

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete read the batch-pipeline brief and define business questions and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Write the business question before code. A reliable pipeline is judged by trusted output and recovery, not by how many libraries it imports.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Toy: "daily valid order value by region" has a business user, freshness need, grain and failure consequence. Write those four before tools.

BUILD THIS ARTIFACT

business_questions.md + assumptions.md

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK P01-01: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-01-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-01/attempt.json.
P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-02.

P01-02 - Profile every source; state grain, keys and defects

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete profile every source; state grain, keys and defects and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Profile each source independently. Grain and business key determine what duplicate means; a repeated customer_id in orders may be normal while repeated order_id may be redelivery.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Compare COUNT rows vs COUNT DISTINCT business key, then inspect nulls/types. Sample one suspected defect manually.

BUILD THIS ARTIFACT

source_profile/ + grain_key_defects.md

VALIDATE

Record row counts, grain, keys, null/duplicate/schema defects; independently sample-check at least one finding.

CHANGED / FAILURE TEST

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-02-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-02/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-03.

P01-03 - Draw architecture and create the repository scaffold

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete draw architecture and create the repository scaffold and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Draw responsibilities, persisted boundaries and failure paths. Raw must be replayable; trusted output must be protected by validation.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Toy flow: CSV/API/DB -> raw batch folder -> validate/quarantine -> trusted -> serving. Mark where state/checkpoint lives.

BUILD THIS ARTIFACT

architecture.md/diagram + ADRs

VALIDATE

Trace every source to trusted/serving output; identify one failure boundary and replay/recovery path.

CHANGED / FAILURE TEST

PHASE CHECK P01-03: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-03-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-03/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-04.

P01-04 - Preserve raw landing evidence and source/run metadata

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete preserve raw landing evidence and source/run metadata and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Landing is evidence, not just a copy. Persist source identity, received time, hash and run status so later replay can prove what arrived.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Two identical business files with different names should be distinguishable as received artifacts but deduplicated by your trusted identity rule.

BUILD THIS ARTIFACT

raw/landing evidence + run_metadata.json

VALIDATE

Source rows/events reconcile to raw + quarantine + deliberate duplicate handling; raw evidence remains replayable.

CHANGED / FAILURE TEST

PHASE CHECK P01-04: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-04-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-04/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-05.

P01-05 - Build ingestion with validation and quarantine

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete build ingestion with validation and quarantine and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Validation decides publish eligibility. Quarantine must preserve rejected record + reason; silent drop is not acceptable evidence.

TINY PATTERN - NOT YOUR FINAL SOLUTION

For one invalid amount and one orphan customer, show exactly which rule fails and where the record goes.

BUILD THIS ARTIFACT

ingestion code + validation/quarantine outputs

VALIDATE

Source rows/events reconcile to raw + quarantine + deliberate duplicate handling; raw evidence remains replayable.

CHANGED / FAILURE TEST

PHASE CHECK P01-05: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-05-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-05/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-06.

P01-06 - Add incremental state and rerun-safe identity

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete add incremental state and rerun-safe identity and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Incremental state is a contract: what was processed, by which key, and when it is safe to advance. Advance only after durable success.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Run the same batch twice. Trusted logical order count/value must not double. Then add O1008 and prove only intended state changes.

BUILD THIS ARTIFACT

state ledger/watermark + rerun/backfill evidence

VALIDATE

Run same input twice and prove trusted logical result is stable; then add a changed/new record and prove only intended state changes.

CHANGED / FAILURE TEST

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-06-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-06/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-07.

P01-07 - Build trusted PostgreSQL/dbt-style transformations

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete build trusted postgresql/dbt-style transformations and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Trusted transformation begins with output grain. Before joining, prove dimension/reference keys are unique at the join grain.

TINY PATTERN - NOT YOUR FINAL SOLUTION

A duplicate customer dimension row can double one order. Detect fan-out with row count/distinct order_id and value reconciliation.

BUILD THIS ARTIFACT

models/SQL/dbt + lineage + row-count controls

VALIDATE

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

CHANGED / FAILURE TEST

Inject one duplicate dimension/reference key or one-to-many fan-out risk and prove the validation catches it.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-07-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-07/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-08.

P01-08 - Add tests and source-to-target reconciliation

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete add tests and source-to-target reconciliation and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Tests must assert behavior and business invariants. Include at least one test that fails on a supplied bad condition.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Separate unit behavior, data-contract tests and source-to-target reconciliation; a green placeholder test proves nothing.

BUILD THIS ARTIFACT

tests/ + reconciliation report + quarantine evidence

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK P01-08: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-08-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-08/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-09.

P01-09 - Automate the repeatable run/orchestration path

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete automate the repeatable run/orchestration path and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Orchestration coordinates persisted work; it should not hide state. A downstream publish must depend on upstream quality success.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Simulate ingest success -> quality failure -> publish blocked. After repair, rerun only safe required steps and reconcile.

BUILD THIS ARTIFACT

DAG/workflow + run log + dependency evidence

VALIDATE

Run dependency path end-to-end; prove failed task does not falsely advance downstream state; rerun only required work.

CHANGED / FAILURE TEST

Force an upstream task failure/partial run; prove downstream publish does not falsely succeed and rerun is bounded.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-09-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-09/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-10.

P01-10 - Inject failure, diagnose, repair and prove recovery

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete inject failure, diagnose, repair and prove recovery and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Failure practice starts by freezing the first failure. Root cause is why the system behaved incorrectly, not the last error line.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Use one sealed failure: reproduce -> save evidence -> diagnose -> smallest repair -> fresh changed retry -> rerun prior controls.

BUILD THIS ARTIFACT

first-failure evidence + root_cause.md + repair + fresh retry evidence

VALIDATE

Save first failing evidence before repair; identify root cause; apply smallest safe repair; fresh retry must pass all prior controls.

CHANGED / FAILURE TEST

MANDATORY SUPPLIED FAILURE: capture first failure, diagnose, repair, fresh retry; do not overwrite original evidence.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-10-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-10/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-11.

P01-11 - Finish README, runbook, architecture and evidence index

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete finish readme, runbook, architecture and evidence index and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Documentation is an operational interface. A new engineer should know how to run, validate, recover and locate evidence.

TINY PATTERN - NOT YOUR FINAL SOLUTION

README = setup/run; runbook = symptoms/recovery; architecture = responsibilities; evidence index = proof paths.

BUILD THIS ARTIFACT

architecture.md/diagram + ADRs

VALIDATE

Trace every source to trusted/serving output; identify one failure boundary and replay/recovery path.

CHANGED / FAILURE TEST

PHASE CHECK P01-11: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-11-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-11/attempt.json. P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P01-12.

P01-12 - Clean-clone rerun and project defense

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete clean-clone rerun and project defense and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

A clean-clone rerun tests hidden dependency risk. Defense tests whether you understand the system without reading code.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Rebuild from documented dependencies/config, run controls, then answer grain, idempotency, failure and one rejected alternative.

BUILD THIS ARTIFACT

state ledger/watermark + rerun/backfill evidence

VALIDATE

Run same input twice and prove trusted logical result is stable; then add a changed/new record and prove only intended state changes.

CHANGED / FAILURE TEST

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P01-12-EVIDENCE. Save: 03_PROJECT_WORKSPACES/P01_RELIABLE_BATCH_PIPELINE/evidence/P01-12/attempt.json.
P01 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P02-01.